

Name: K RAJ BHARATH K RAJ BHARATH | Reg. No: RA2511003011548

Unit – 5 : Session – 1 : SLO – 2 | Types of Graphs

■ I. Definitions

- a) Simple Graph: an undirected graph with no self-loops and no multiple edges between the same pair of vertices.
- b) Directed Graph (digraph): a graph in which every edge has a direction, going from one vertex to another.
- c) Weighted Graph: a graph in which each edge carries a value (weight) such as cost, distance or time.
- d) Complete Graph: a graph in which every pair of distinct vertices is connected by an edge ($n(n-1)/2$ edges).
- e) Null Graph: a graph with vertices but no edges.
- f) Connected Graph: a graph in which a path exists between every pair of vertices.
- g) Disconnected Graph: a graph in which at least one pair of vertices has no path between them.
- h) Multigraph: a graph that allows multiple (parallel) edges between the same pair of vertices.
- i) Subgraph: a graph whose vertices and edges are subsets of another graph's vertices and edges.
- j) Cyclic Graph: a graph that contains at least one cycle (a path that starts and ends at the same vertex).
- k) Undirected Graph: a graph in which edges have no direction; edge (u, v) is the same as (v, u).

■ II. Diagrams for Three Graph Types

- Simple graph: A---B (plain edges, no loops)
- ||
- C---D
- Directed graph: A -> B -> C -> A (arrows show direction of edges)
- Weighted graph: A --5-- B
- ||
- 2 3
- ||
- C --4-- D (numbers on edges are weights)

■ III. True or False

- a) A complete graph with 'n' vertices has $n(n-1)/2$ edges — TRUE.
- b) In a weighted graph, no values are assigned to edges — FALSE.
- c) A directed graph has edges with direction — TRUE.
- d) A multigraph allows multiple edges between two vertices — TRUE.

■ IV. Match the Following

- a) Weighted Graph — ii) Values on edges
- b) Null Graph — i) No edges
- c) Directed Graph — iv) Arrows showing direction
- d) Multigraph — iii) Loops or multiple edges

■ V. Real-Life Applications

- Weighted graph: road networks in Google Maps (edge weight = distance/travel time) for shortest-path routing.
- Directed graph: Twitter/Instagram 'follow' relationships (one-way edges), or web links between pages.
- Complete graph: round-robin tournament scheduling where every team plays every other team.
- Null graph: a set of newly added users in a network with no connections yet.